Therapeutic Drug Monitoring (TDM)

Principles, pharmacokinetic calculations, and every drug that needs TDM — therapeutic ranges, sampling times, toxicity, and key factors

1. Principles of TDM

- **TDM** = measuring drug concentrations (usually serum/plasma; **whole blood** for tacrolimus, cyclosporine, sirolimus, everolimus) and using pharmacokinetic (PK) principles to **individualize dosing** → maximize efficacy and minimize toxicity.
- **The level must always be interpreted with the clinical picture** ("treat the patient, not the level") — a "therapeutic" level can still be toxic or ineffective in some patients.

Criteria — a drug is a good candidate for TDM when

1. **Narrow therapeutic index** (toxic concentration close to effective concentration).
2. **Large inter-patient PK variability** (same dose → very different levels) — genetics, organ function, DDIs, nonlinear kinetics.
3. **Good relationship between concentration and effect/toxicity** (better than dose–response).
4. **No easily measurable clinical endpoint** for efficacy (e.g., seizures prevented, transplant rejection, infection cure) — drugs with easy endpoints (BP, glucose, INR) are monitored by the endpoint, not level.
5. **Toxicity is serious or hard to distinguish** from the disease (e.g., digoxin toxicity vs HF symptoms, phenytoin toxicity vs seizures).
6. A **reliable, available, and timely assay** exists.

When to order a level (indications)

- **After starting therapy or a dose change** (at steady state) • **suspected toxicity** • **lack of efficacy** (therapeutic failure) • suspected **non-adherence** • **changing organ function** (renal, hepatic failure, dialysis, critical illness) • **drug interactions** (adding/removing inducers or inhibitors) • **pregnancy** (↑ clearance and Vd) • changes in **protein binding** (hypoalbuminemia) • changes in dosage form/route or manufacturer • extremes of age/weight • smoking status changes.

Pharmacokinetic basics (high-yield formulas)

Parameter	Formula / concept
Half-life ($t_{1/2}$)	**$t_{1/2} = 0.693 \times Vd / CL = 0.693 / ke$** • time for concentration to fall by 50%
Elimination rate constant (ke)	**$ke = \ln(C1/C2) / \Delta t$** (two levels in the elimination phase) = CL/Vd
Steady state (Css)	Reached after **4–5 half-lives** (94–97%) — independent of dose • a **loading dose** achieves target levels immediately
Loading dose	**$LD = (Vd \times Ctarget) / (S \times F)$** • S = salt factor, F = bioavailability
Maintenance dose	**$MD = (CL \times Css,avg \times \tau) / (S \times F)$** • τ = dosing interval
Linear (first-order) kinetics	Most drugs: Css changes **proportionally** with dose → **new dose = old dose × (target level / current level)**
Nonlinear (Michaelis–Menten, zero-order/saturable)	**Phenytoin** (also high-dose aspirin, ethanol, valproate protein binding): small dose ↑ → **disproportionately large** level ↑ and longer $t_{1/2}$
Creatinine clearance (Cockcroft–Gault)	**$CrCl = [(140 - age) \times weight] / (72 \times SCr)$** (× 0.85 for females)
Ideal body weight (IBW)	Men **50 kg** + 2.3 kg per inch over 5 ft • women **45.5 kg** + 2.3 kg per inch over 5 ft
Adjusted body weight (AdjBW)	**$AdjBW = IBW + 0.4 \times (TBW - IBW)$** — used for aminoglycosides (and CrCl) when TBW > 120–130% of IBW

Sampling — getting it right

- **Trough** = lowest concentration, drawn **just before the next dose** (within ~30 min before) — most commonly used (efficacy/accumulation) • **peak** = drawn after absorption/distribution complete (e.g., 30 min after end of a 30-min aminoglycoside infusion) — efficacy for concentration-dependent drugs.
- **Draw at steady state** (after 4–5 t½) unless toxicity suspected or loading dose given • record **exact time of dose and sample**.
- **Distribution phase**: avoid sampling too early — **digoxin ≥ 6–8 h after dose**, **lithium 12 h after dose**.
- **Don't draw from the IV line** used to infuse the drug (falsely high) • correct tube type (no gel separator for some drugs) • whole blood (EDTA) for calcineurin/mTOR inhibitors.
- **Total vs free (unbound) level:** only free drug is active • highly protein-bound drugs (**phenytoin** ~90%, **valproate** ~90%, carbamazepine) → in **hypoalbuminemia, renal failure (uremia), pregnancy, elderly, critical illness, displacing drugs** the total level underestimates activity → **measure free level** or correct.
- **Common reasons for unexpected levels:** wrong sample timing, non-adherence, DDIs, organ dysfunction, lab error, dosing error, drug not at steady state, formulation change.

2. Aminoglycosides (gentamicin, tobramycin, amikacin)

- **Concentration-dependent killing** (peak/MIC ≥ 8–10) + **post-antibiotic effect** → efficacy linked to **peak** • **toxicity** (nephrotoxicity, ototoxicity) linked to **elevated trough**/prolonged exposure.
- **Renally eliminated** (glomerular filtration) • distribute in extracellular fluid (Vd ~0.25 L/kg; ↑ in edema, ascites, burns, sepsis; ↓ relative in obesity).
- **Dosing weight: IBW** (or TBW if less than IBW) • **AdjBW** if TBW > 120% IBW (obesity).

Regimen	Gentamicin / tobramycin	Amikacin
Traditional (conventional) dosing	1–2.5 mg/kg q8h (adjusted by CrCl) • **peak 5–10 mcg/mL** (UTI 4–6; pneumonia/sepsis **8–10**) • **trough < 1–2 mcg/mL** (goal < 1)	5–7.5 mg/kg q8–12h • **peak 20–35 mcg/mL** • **trough < 4–8 mcg/mL** (goal < 5)
Extended-interval (once-daily) dosing	**5–7 mg/kg q24h** (interval extended by CrCl) • **random level 6–14 h after start** plotted on the **Hartford nomogram** (7 mg/kg) → determines interval (q24/q36/q48h) • **trough undetectable (< 0.5–1)** • peak ~16–24	**15–20 mg/kg q24h** • trough < 1–4 (undetectable) • peak ~50–60
Synergy (gram-positive endocarditis)	**1 mg/kg q8h** or 3 mg/kg daily (strep) • **peak 3–4 mcg/mL** • **trough < 1**	—
Cystic fibrosis (tobramycin)	10 mg/kg once daily (high Vd and CL) • peak 20–30 • trough < 1	—

- **Sampling: peak 30 min after end of a 30-min infusion** (1 h after start; 1 h after IM) • **trough within 30 min before the next dose** • first levels around the **3rd–4th dose** (steady state) in conventional dosing; random level after the 1st dose for extended-interval.
- **Individualized dosing (Sawchuk–Zaske):** from peak and trough → **ke = ln(Cpeak/Ctrough)/Δt** → t½ = 0.693/ke → Vd → calculate new dose and interval to reach target peak/trough.
- **Avoid extended-interval dosing** in: **pregnancy**, **burns > 20%**, **ascites**, **CrCl < 20–30**/dialysis, **synergy for endocarditis**, children, mycobacterial infections, enterococcal infections, cystic fibrosis (different protocol).
- **Toxicity: nephrotoxicity** (non-oliguric AKI after 5–7 days, usually reversible; risk ↑ with trough elevation, duration > 7–10 days, vancomycin, amphotericin, loop diuretics, cisplatin, cyclosporine, NSAIDs, contrast, volume depletion) • **ototoxicity** (vestibular — vertigo, ataxia; cochlear — hearing loss/tinnitus; often **irreversible**) • **neuromuscular blockade** (myasthenia gravis, with NMBs).
- **Monitor:** SCr every 2–3 days (daily if unstable), urine output, levels, hearing/vestibular symptoms • limit duration.
- **Hemodialysis:** dose after dialysis, check pre-dialysis levels.

3. Vancomycin (ASHP/IDSA 2020 consensus)

- **AUC-dependent** killing (AUC24/MIC) • **target AUC24/MIC 400–600 mg·h/L** (assuming MIC 1) for **serious MRSA infections** (bacteremia, endocarditis, osteomyelitis, meningitis, pneumonia) → efficacy with **less nephrotoxicity** than trough-guided dosing.
- **Trough-only monitoring (15–20 mg/L) is no longer recommended** (troughs > 15 ↑ AKI without better efficacy) — but still used in many centers/exams.
- **AUC estimation: Bayesian software** (preferred — can use 1 level, even before steady state) **or two levels** (peak 1–2 h after end of infusion + trough) with first-order equations (trapezoid) • **continuous infusion: AUC24 = Css × 24** (target Css 15–25 mg/L).
- **Dosing: loading dose 20–35 mg/kg** (actual body weight; max 3,000 mg) in serious infections/critically ill • **maintenance 15–20 mg/kg q8–12h** (by renal function) • obesity: load 20–25 mg/kg, maintenance ≤ 4.5 g/day initially • infuse ≤ 1 g/h (vancomycin infusion reaction).
- **Monitoring:** levels within **24–48 h** (early for serious infections) • **SCr at least every 2–3 days** (daily if unstable) • TDM may not be needed for short courses (< 3–5 days) or uncomplicated SSTI with stable renal function.
- **Historical trough targets: 10–15 mg/L** (uncomplicated) • **15–20 mg/L** (serious infections — meningitis, endocarditis, osteomyelitis, pneumonia, bacteremia) • always > 10 to avoid resistance.
- **Toxicity: AKI** (↑ with AUC > 600, trough > 15–20, **piperacillin-tazobactam**, aminoglycosides, loop diuretics, contrast, ICU) • **vancomycin infusion reaction** ("red man" — histamine; slow infusion, antihistamine) • ototoxicity (rare) • neutropenia, thrombocytopenia (prolonged) • DRESS.
- **Pediatrics:** AUC 400–600 (60–80 mg/kg/day in divided doses) • **hemodialysis:** pre-dialysis levels 15–20.
- **Oral vancomycin** (C. difficile) is not absorbed → **no TDM**.

4. Digoxin

- **Therapeutic range: HF: 0.5–0.9 ng/mL** (higher levels ↑ mortality — DIG trial) • **AF rate control: 0.5–2.0 ng/mL** (many recommend ≤ 1.2) • toxicity more likely > 2 ng/mL (can occur at lower levels with risk factors).
- **Sampling: at least 6–8 h after the dose** (ideally 12–24 h / just before next dose) — long **distribution phase** (levels drawn earlier are falsely high) • **steady state 7–10 days** (t½ ~36–48 h) — **up to 2–3 weeks in renal failure** (t½ ↑ to 4–6 days).
- **Pharmacokinetics: renal elimination** (~70% unchanged) → ↓ dose in CKD/elderly • large Vd (~7 L/kg; ↓ in renal failure and hypothyroidism) • bioavailability tablets 70–80%, IV 100% • not removed by dialysis.
- **Toxicity:** GI (**anorexia, nausea, vomiting**, abdominal pain) • CNS (fatigue, confusion, delirium) • **visual** (**yellow-green halos**, blurred vision) • **arrhythmias** (PVCs, bigeminy, **bidirectional VT**, atrial tachycardia with block, AV block, bradycardia, VF) • **hyperkalemia in acute overdose** (poor prognosis).
- **Factors ↑ toxicity (even with normal levels): hypokalemia**, **hypomagnesemia**, **hypercalcemia**, hypothyroidism, hypoxia, ischemia, acidosis, elderly.
- **DDIs ↑ digoxin: amiodarone** (↓ digoxin 50%), **dronedarone** (↓ 50%), **verapamil**, diltiazem, **quinidine**, propafenone, **clarithromycin/erythromycin**, itraconazole/ketoconazole, cyclosporine, ritonavir, ranolazine, ticagrelor, carvedilol, spironolactone (also assay interference) • **↓ digoxin:** rifampin, St John's wort, cholestyramine/colestipol, antacids, kaolin, sucralfate • diuretics → hypokalemia.
- **Digoxin immune Fab (DigiFab)** for life-threatening toxicity • dose (vials) = **(serum digoxin level × weight in kg) / 100** (chronic) or **(mg ingested × 0.8) / 0.5** (acute) • **after Fab, total digoxin levels are falsely ↑ and unreliable** (measure free digoxin) • rebound in renal failure.
- **Digoxin-like immunoreactive substances (DLIS)** → falsely ↑ levels in **renal failure, liver disease, pregnancy, neonates**.

5. Antiepileptic Drugs (AEDs)

Phenytoin (most tested TDM drug)

- **Therapeutic range: total 10–20 mcg/mL** (mg/L) • **free (unbound) 1–2 mcg/mL** (normally ~10% free).
- **Nonlinear (Michaelis–Menten) kinetics** — saturable hepatic metabolism (**CYP2C9**, 2C19) → **small dose changes cause large level changes**; t½ increases with dose • practical rule for dose increases: level **< 7** → ↑ by **100 mg/day**; **7–12** → ↑ by **50 mg/day**; **> 12** → ↑ by **≤ 30 mg/day** • steady state takes **5–7 days to several weeks** (longer at higher levels).
- **Highly protein bound (~90%) to albumin** → **hypoalbuminemia** (< 3.5 g/dL) or **renal failure** ↑ free fraction → **total level underestimates** active drug → use **free level** (best) or correct the total level.
- **Albumin correction (Sheiner–Tozer / Winter–Tozer): C(corrected) = C(measured) / [(0.2 × albumin) + 0.1] • if CrCl < 10 mL/min (ESRD): C / [(0.1 × albumin) + 0.1]**.
- **Concentration-related toxicity: > 20 → nystagmus** (lateral gaze) • **> 30 → ataxia**, slurred speech, diplopia • **> 40 → lethargy, confusion**, ↓ cognition • very high → coma, paradoxical seizures.
- **Non-concentration-related ADRs: gingival hyperplasia**, **hirsutism**, coarsening of facial features, acne, **rash/SJS/TEN** (**HLA-B*1502** in Asians), osteomalacia (↓ vitamin D), megaloblastic anemia (folate), hepatotoxicity, lymphadenopathy, teratogenic (fetal hydantoin syndrome), **purple glove syndrome** (IV).
- **Loading dose: 15–20 mg/kg IV** at ≤ **50 mg/min** (≤ 25 mg/min elderly/CV disease) — **hypotension, arrhythmias** (propylene glycol) → cardiac monitoring • mix only in **normal saline** (precipitates in dextrose) • **fosphenytoin** (prodrug, dosed in **PE**) up to **150 mg PE/min**, fewer infusion reactions, IM possible.
- **Sampling:** trough at steady state • after IV loading → ~**2 h after phenytoin infusion** (fosphenytoin: **≥ 2 h after IV, ≥ 4 h after IM** for conversion).
- **Formulations:** phenytoin **sodium** (capsules, IV) = **92% phenytoin** (S = 0.92) • phenytoin **acid** (chewable tablets, suspension) = 100% → switching forms changes the dose • **shake suspension** well • **enteral feeds ↓ absorption** → hold feeds 1–2 h before and after.
- **DDIs:** phenytoin is a **potent inducer** (CYP3A4, 2C9, 2C19, UGT) → ↓ OCs, warfarin (complex), DOACs, azoles, many drugs • **↑ phenytoin: amiodarone**, **fluconazole**, isoniazid, **valproate** (displacement + inhibition → ↑ free), metronidazole, TMP-SMX, omeprazole, cimetidine, fluoxetine • **↓ phenytoin:** carbamazepine, rifampin, chronic alcohol, folic acid, enteral feeds, antacids/calcium.

Other antiepileptics

Drug	Therapeutic range	Key TDM points
Carbamazepine	**4–12 mcg/mL**	**Autoinduction** (induces its own metabolism over **2–4 weeks**) → levels fall → **recheck after 3–4 weeks** and titrate • active **10,11-epoxide** metabolite (toxicity) • CYP3A4 substrate: **↑ by macrolides, azoles, grapefruit, diltiazem/verapamil, fluoxetine, isoniazid**; ↓ by phenytoin, rifampin • toxicity: **diplopia**, dizziness, ataxia, nausea • ADRs: **hyponatremia/SIADH**, aplastic anemia/agranulocytosis (CBC), **SJS/TEN (HLA-B*1502)**, hepatotoxicity
Valproic acid	**Epilepsy 50–100 mcg/mL** • mania 50–125 • free 5–15	**Saturable protein binding** (free fraction ↑ at high levels and in hypoalbuminemia, uremia, elderly) → total level misleading → measure **free level** • trough sampling • ADRs: **hepatotoxicity (boxed)**, **pancreatitis (boxed)**, **teratogenic (boxed — neural tube defects)**, **thrombocytopenia** (↑ at levels > 110 in women/> 135 in men), **hyperammonemia** (check ammonia; L-carnitine), weight gain, tremor, alopecia • **carbapenems ↓ VPA levels drastically** (avoid) • VPA ↑ lamotrigine (rash), phenobarbital; inhibits phenytoin metabolism
Phenobarbital	**15–40 mcg/mL** (neonates 15–40)	Very long t½ (~100 h; 2–5 days) → **steady state 2–3 weeks** • potent inducer • sedation, cognitive/behavioral effects, respiratory depression at high levels • ↑ by valproate
Primidone	5–12 mcg/mL (+ phenobarbital metabolite 15–40)	Measure both primidone and phenobarbital
Ethosuximide	40–100 mcg/mL	Absence seizures • GI, hiccups
Lamotrigine	~3–15 mcg/mL (reference range; not routinely needed)	TDM in **pregnancy** (clearance ↑ up to 2–3×), with **estrogen OCs** (↓ levels), **valproate** (↑ levels — halve dose), inducers (↓) • rash/SJS with rapid titration
Levetiracetam	~12–46 mcg/mL (reference)	Not routine • renal elimination • pregnancy/adherence
Oxcarbazepine	MHD metabolite 3–35 mcg/mL	Hyponatremia • pregnancy
Others (topiramate 5–20, zonisamide 10–40, lacosamide ~10–20, felbamate 30–60)	Reference ranges	Not routinely monitored — used for adherence, toxicity, pregnancy, DDIs, renal/hepatic impairment

- **Pregnancy** ↑ clearance of most AEDs (lamotrigine, levetiracetam, oxcarbazepine, phenytoin) → monitor levels (baseline pre-pregnancy, each trimester, postpartum — reduce dose after delivery).

6. Psychiatric Drugs

Lithium

- **Therapeutic range: acute mania 0.8–1.2 mEq/L** (up to 1.5) • **maintenance 0.6–1.0 mEq/L** (0.8–1.0 optimal) • **elderly 0.4–0.8**.
- **Sampling: 12 h after the last dose** (morning trough, before the morning dose) • **steady state in ~5 days** (t½ ~24 h) → check **5 days after starting or changing dose** (more frequently initially), then every **3–6 months** when stable.
- **Pharmacokinetics:** not metabolized, **100% renal** excretion — handled like **sodium** (reabsorbed in proximal tubule) → **sodium depletion or dehydration ↑ lithium reabsorption → toxicity** • not protein bound • dialyzable.
- **Toxicity: > 1.5 (mild)**: N/V/D, **coarse tremor**, weakness, sedation • **> 2.0 (moderate)**: confusion, **ataxia**, dysarthria, nystagmus, muscle twitching, hyperreflexia • **> 2.5–3.0 (severe)**: seizures, coma, arrhythmias, renal failure, death → **hemodialysis** (level > 4, or > 2.5 with severe symptoms/renal failure) • no activated charcoal (doesn't bind) — whole bowel irrigation for SR.

- ↑ **Lithium levels: thiazide diuretics** (↑ 25–40%), **ACE inhibitors**, **ARBs**, **NSAIDs** (except aspirin, sulindac), **dehydration**, **low-sodium diet**, vomiting/diarrhea, fever/sweating, **renal impairment**, elderly, metronidazole, tetracyclines, SGLT2i (variable) • ↓ **Lithium levels: caffeine**, **theophylline**, high sodium intake, **pregnancy** (↑ GFR), osmotic diuretics, acetazolamide, sodium bicarbonate.
- **Other monitoring: SCr/eGFR** and **TSH** (hypothyroidism, goiter) every 6–12 months • **calcium** (hyperparathyroidism) • electrolytes • ECG (age > 40, cardiac) • weight • pregnancy test (**Ebstein's anomaly**) • ADRs: **nephrogenic diabetes insipidus** (polyuria/polydipsia — amiloride), fine tremor (propranolol), weight gain, acne, psoriasis, leukocytosis, chronic kidney disease.

Clozapine

- **Target clozapine level ≥ 350 ng/mL** (therapeutic response; range ~350–600) • check if non-response, toxicity, or smoking/DDI changes • trough (12 h post dose) at steady state.
- **CYP1A2 substrate → smoking (polycyclic hydrocarbons) induces 1A2 → smoking cessation ↑ clozapine levels by ~50%** (toxicity, seizures) → ↓ dose ~30–50% and monitor • **fluvoxamine**, **ciprofloxacin**, cimetidine, oral contraceptives ↑ levels • carbamazepine/rifampin ↓ levels (carbamazepine also ↑ agranulocytosis risk).
- **Toxicity: seizures** (dose/level-dependent, ↑ > 600–1,000 ng/mL), sedation, hypotension, delirium, anticholinergic effects.
- **Absolute neutrophil count (ANC) monitoring** for **agranulocytosis** (weekly × 6 months → every 2 weeks × 6 months → monthly) — REMS removed in 2025 but ANC monitoring still recommended clinically • myocarditis (troponin/CRP first month), constipation/ileus, metabolic syndrome, sialorrhea.

Tricyclic antidepressants (TCAs)

TCA	Therapeutic range (parent + active metabolite)
Nortriptyline	**50–150 ng/mL** — **therapeutic window** (levels above the range may be **less effective**)
Amitriptyline (+ nortriptyline)	~80–200 ng/mL (combined)
Imipramine (+ desipramine)	~175–350 ng/mL (combined)
Desipramine	~100–300 ng/mL
Clomipramine, doxepin	Reference ranges exist

- **Indications:** poor response, toxicity, elderly, **CYP2D6 poor/ultrarapid metabolizers**, DDIs (**fluoxetine, paroxetine, bupropion** — 2D6 inhibitors ↑ TCA levels).
- **Toxicity/overdose** (narrow index — 1-week supply can be fatal): **anticholinergic** effects, **sedation**, **orthostatic hypotension**, **QRS widening > 100 ms** (Na-channel blockade → arrhythmias; **> 160 ms → VT**), seizures (QRS > 100), coma → **sodium bicarbonate** • ECG more useful than levels in overdose.

7. Theophylline / Aminophylline

- **Therapeutic range:** classic **10–20 mcg/mL** → now **5–15 mcg/mL** recommended for asthma/COPD (often target **8–12**) • **neonatal apnea 5–15** (caffeine preferred — range 8–20/5–25, rarely monitored).
- **Aminophylline** = theophylline ethylenediamine salt → **80% theophylline** (**S = 0.8**): theophylline dose = aminophylline dose × 0.8.
- **Pharmacokinetics:** hepatic **CYP1A2** (and 3A4, 2E1) • t½ ~8 h in non-smoking adults (shorter in smokers ~4–5 h and children; longer in elderly, HF, liver disease, neonates ~24–30 h) • possible **nonlinear kinetics** at higher levels • sample: trough (or 4–6 h after SR dose; 30 min after IV load; during continuous infusion any time at steady state).
- **Toxicity: > 20**: nausea, vomiting, diarrhea, headache, insomnia, nervousness, **tachycardia**, tremor • **> 30–40**: **seizures**, **arrhythmias** (SVT, VT), hypokalemia, hyperglycemia, metabolic acidosis • seizures may occur without preceding GI symptoms • severe → **multiple-dose activated charcoal**, hemodialysis.
- ↑ **Theophylline levels (↓ clearance): ciprofloxacin**, **erythromycin/clarithromycin**, **cimetidine**, **fluvoxamine**, **zileuton**, allopurinol (high dose), propranolol, verapamil, **oral contraceptives**, interferon, **smoking cessation**, **HF**, **cirrhosis**, **fever/viral illness**, hypothyroidism, elderly, obesity, high-carbohydrate diet.

- ↓ **Theophylline levels (↑ clearance): smoking** (tobacco, marijuana — 1A2 induction), **rifampin**, **phenytoin**, **carbamazepine**, **phenobarbital**, St John's wort, charbroiled meat, high-protein diet, hyperthyroidism, cystic fibrosis, children.
- Theophylline ↓ **lithium** levels and ↓ **adenosine** effect.

8. Immunosuppressants (Transplant)

- Narrow therapeutic index + huge PK variability (absorption, CYP3A4/P-gp, genetics — CYP3A5 expressers need higher tacrolimus doses) + serious consequences (**rejection** vs **toxicity**) → **routine TDM**.
- **Whole blood** (EDTA tube) for **tacrolimus, cyclosporine, sirolimus, everolimus** (extensive RBC binding) • **trough (C0)** just before the next dose (12 h for BID) • targets depend on **organ**, **time since transplant** (higher early), assay, and combination regimen.

Drug	Usual trough target	Key toxicities / notes
Tacrolimus (calcineurin inhibitor)	**~5–15 ng/mL** (e.g., kidney: 8–12 early → 5–8 later; liver 5–10)	**Nephrotoxicity**, **neurotoxicity** (tremor, headache, seizures, PRES), **hyperglycemia/new-onset diabetes**, **hyperkalemia**, **hypomagnesemia**, HTN, QT prolongation, alopecia • take consistently with or without food • IR (BID) vs ER (daily; Envarsus, Astagraf) not interchangeable
Cyclosporine (calcineurin inhibitor)	**~100–400 ng/mL** (C0; depends on organ/time) • **C2 monitoring** (2 h post-dose) alternative	**Nephrotoxicity**, **HTN**, **hyperlipidemia**, **gingival hyperplasia**, **hirsutism**, tremor, hyperkalemia, hypomagnesemia, hyperuricemia/gout • **Neoral/Gengraf (modified) ≠ Sandimmune** (not interchangeable) • CYP3A4 inhibitor itself (↑ statins)
Sirolimus (mTOR inhibitor)	**~5–15 ng/mL** (4–12)	Long t½ (~60 h) → steady state ~1–2 weeks • **hyperlipidemia**, **myelosuppression**, **impaired wound healing**, mouth ulcers, pneumonitis, proteinuria, edema • less nephrotoxic • CI with voriconazole/posaconazole
Everolimus	**~3–8 ng/mL** (transplant)	Similar to sirolimus
Mycophenolate (MPA)	AUC 30–60 mg·h/L (not routine)	GI (diarrhea), leukopenia, teratogenic (REMS) • usually not monitored

Calcineurin/mTOR inhibitor DDIs (CYP3A4 and P-gp substrates): ↑ levels (toxicity): azoles (ketoconazole, itraconazole, **voriconazole** — ↓ tacrolimus dose by 2/3, posaconazole, fluconazole), **macrolides** (clarithromycin, erythromycin), **diltiazem**, **verapamil**, nicardipine, **protease inhibitors**/cobicistat (huge ↑), amiodarone, metoclopramide, **grapefruit juice • ↓ levels (rejection): rifampin**, rifabutin, **carbamazepine**, **phenytoin**, phenobarbital, **St John's wort**, nafcillin, efavirenz • additive nephrotoxicity: aminoglycosides, amphotericin B, NSAIDs, vancomycin • hyperkalemia: ACEi/ARB, K-sparing diuretics, TMP-SMX.

9. Antiarrhythmics (TDM)

Drug	Therapeutic range	Key points
Lidocaine	**1.5–5 mcg/mL**	Hepatic (high extraction — ↓ with HF, liver disease, ↓ hepatic blood flow, β-blockers) • active metabolites (MEGX) • toxicity > 5–6: **CNS** (perioral numbness, tinnitus, confusion, slurred speech, **seizures > 8**), bradycardia • lipid emulsion for LAST
Procainamide + **NAPA**	Procainamide **4–10 mcg/mL** • NAPA **15–25 mcg/mL** • (sum 10–30)	Acetylated (NAT2) to **NAPA** (class III, **renally cleared** — accumulates in CKD) • **fast acetylators** → more NAPA • toxicity: QRS/QT prolongation, torsades, hypotension • lupus (slow acetylators), agranulocytosis
Quinidine	2–5 mcg/mL	Cinchonism, diarrhea, QT, ↑ digoxin

Drug	Therapeutic range	Key points
Flecainide	0.2–1.0 mcg/mL	Proarrhythmia, CNS • CYP2D6
Amiodarone (+ desethylamiodarone)	~1.0–2.5 mcg/mL	**Not routinely monitored** (poor correlation) — clinical and organ toxicity monitoring (TFTs, LFTs, CXR)
Digoxin	0.5–0.9 (HF) / ≤ 2 (AF) ng/mL	See Section 4

10. Antimicrobials Other Than Aminoglycosides and Vancomycin

Drug	Target	Key points
Voriconazole	**Trough 1–5.5 mcg/mL** (some 2–5 for severe)	**CYP2C19 polymorphism** + nonlinear kinetics → huge variability • check trough after **5–7 days** • < 1 → failure • **> 5.5 → neurotoxicity** (visual hallucinations, encephalopathy) and **hepatotoxicity** • DDIs (strong 3A4 inhibitor)
Posaconazole	Prophylaxis trough **> 0.7 mcg/mL** • treatment **> 1–1.25**	Absorption variable (suspension needs food; DR tablets better) • check after ~5–7 days • PPIs ↓ suspension absorption
Itraconazole	Trough **> 0.5–1 mcg/mL** (itraconazole + hydroxy-itraconazole > 1–2)	Erratic absorption (capsule needs acid/food) • check after ~2 weeks
Flucytosine	**Peak 30–80 mcg/mL** (2 h post-dose) • trough 25–50	**> 100 → bone marrow suppression**, hepatotoxicity • renal elimination (adjust in CKD, with amphotericin-induced AKI)
Isavuconazole, fluconazole	Not routinely monitored	Fluconazole linear, predictable
Linezolid	Trough ~2–7 mg/L (emerging)	Trough > 7–8 → **thrombocytopenia** (renal impairment, prolonged therapy)
β-lactams (ICU)	% time free drug above MIC (100% fT>MIC; emerging)	Critically ill, augmented renal clearance, CRRT, ECMO, obesity • cefepime neurotoxicity
Polymyxins (colistin, polymyxin B)	Colistin Css ~2 mg/L (AUC-based)	Narrow index, nephrotoxicity, neurotoxicity
Chloramphenicol	Peak 10–25 mcg/mL	> 25 → dose-related marrow suppression • **gray baby syndrome** in neonates • idiosyncratic aplastic anemia
Anti-TB drugs	INH peak 3–6 mcg/mL (2 h) • RIF peak 8–24 mcg/mL (2 h)	For slow response, malabsorption, HIV, diabetes, MDR-TB (not routine)
Antiretrovirals	Not routine	Pregnancy, DDIs, pediatrics in selected cases

11. Anticoagulants (Pharmacodynamic/Level Monitoring)

Drug	Monitoring test / target	Notes
Warfarin	**INR 2.0–3.0** (most indications) • **2.5–3.5** mechanical mitral valve (or aortic + risk factors)	INR every 2–3 days initially → up to every 12 weeks when stable • many DDIs (CYP2C9 — amiodarone, fluconazole, metronidazole, SMX-TMP ↑; rifampin, carbamazepine ↓) • vitamin K diet consistency
Unfractionated heparin	**aPTT 1.5–2.5 × control** or **anti-Xa 0.3–0.7 units/mL**	6 h after start/dose change • ACT during PCI/cardiac surgery • platelets for HIT
LMWH (enoxaparin)	**Anti-Xa peak 4 h after 3rd–4th dose**: **0.5–1.0** (q12h) • 1.0–2.0 (daily) • prophylaxis 0.2–0.5	Only in **renal impairment**, **obesity**, **pregnancy**, low body weight, children
Argatroban / bivalirudin	**aPTT 1.5–3 × baseline**	Argatroban falsely ↑ INR

Drug	Monitoring test / target	Notes
Fondaparinux	Drug-specific anti-Xa (rarely)	—
DOACs	Not routinely monitored	Drug-specific anti-Xa or dilute thrombin time for emergencies (bleeding, surgery, overdose, extremes of weight)

12. Oncology, Biologics and Others

High-dose methotrexate (HDMTX)

- **Plasma MTX levels at 24, 48, and 72 h** after start of infusion • typical thresholds: **< 10 µmol/L at 24 h**, **< 1 µmol/L at 48 h**, **< 0.1 µmol/L at 72 h** → **leucovorin (folinic acid) rescue** continued until level **< 0.05–0.1 µmol/L** (dose ↑ if delayed clearance).
- **Prevent toxicity: aggressive IV hydration** + **urine alkalinization (sodium bicarbonate, urine pH ≥ 7–8)** (MTX precipitates in acidic urine → AKI) • monitor SCr, urine pH, urine output • **glucarpidase** (carboxypeptidase G2) for toxic levels with renal dysfunction.
- **Toxicity: nephrotoxicity**, **myelosuppression**, **mucositis**, hepatotoxicity, neurotoxicity.
- **DDIs ↓ MTX clearance: NSAIDs**, **penicillins**, **PPIs**, **TMP-SMX** (also folate antagonist), probenecid, salicylates, ciprofloxacin • third-spacing (**ascites, pleural effusion**) → prolonged elimination (drain before HDMTX).
- Low-dose weekly MTX (RA/psoriasis): monitor CBC, LFTs, SCr — not levels.

Other oncology agents

- **Busulfan** (IV, HSCT conditioning) → **AUC-targeted dosing** (narrow index; ↑ AUC → veno-occlusive disease, seizures — give seizure prophylaxis; ↓ AUC → graft failure/relapse).
- **5-fluorouracil** (AUC-guided dosing in some centers) • **mitotane** 14–20 mg/L • **imatinib** trough ≥ 1,000 ng/mL (selected) • carboplatin dosed by **Calvert formula** (dose = target AUC × (GFR + 25)) — not a level but AUC-based.

Biologics (proactive/reactive TDM in IBD)

- **Infliximab trough ≥ 5 mcg/mL** (3–7; ≥ 10 for fistulizing disease) • **adalimumab trough ≥ 7.5 mcg/mL** (5–12) • measure with **anti-drug antibodies**: **low level + no antibodies** → dose intensification • **low level + high-titer antibodies** → switch within class (or add immunomodulator) • **therapeutic level + active disease** → switch mechanism (out of class) • AGA supports **reactive TDM** in active disease on anti-TNF.

Toxicology levels (overdose)

Drug	Level / interpretation
Acetaminophen	**Rumack–Matthew nomogram**: plot level drawn **4–24 h** after a single acute ingestion • **treatment line starts at 150 mcg/mL at 4 h** → **N-acetylcysteine** • levels before 4 h are uninterpretable • not valid for chronic/staggered or unknown time ingestion (treat if detectable level or ↑ ALT) • extended-release: repeat level 4–6 h later
Salicylates	Anti-inflammatory 15–30 mg/dL (150–300 mcg/mL) • **toxicity > 30–40 mg/dL** (tinnitus, N/V, respiratory alkalosis + AG metabolic acidosis) • **> 90–100 mg/dL (acute) or > 60 (chronic) → hemodialysis** • repeat levels every 2 h (delayed absorption) • Done nomogram obsolete
Iron	Serum iron 4–6 h post-ingestion: **> 350 mcg/dL** moderate • **> 500 mcg/dL** severe → **deferoxamine**
Lithium, digoxin, theophylline, phenytoin, carbamazepine, valproate, TCAs	See individual sections
Methanol / ethylene glycol	Levels > 20 mg/dL → fomepizole ± dialysis • osmolal gap
Ethanol	Legal intoxication ≥ 80 mg/dL (0.08%)

13. Factors That Alter Drug Levels (apply to all TDM drugs)

Factor	Effect
Renal impairment	↑ levels of renally cleared drugs: **aminoglycosides, vancomycin, digoxin, lithium, NAPA, flucytosine, enoxaparin, levetiracetam** • ↓ protein binding (phenytoin, valproate — ↑ free fraction) • DLIS (digoxin assay)
Hepatic impairment	↑ levels of hepatically cleared drugs: **phenytoin, carbamazepine, valproate, theophylline, lidocaine, tacrolimus, cyclosporine, voriconazole** • ↓ albumin → ↑ free fraction
Hypoalbuminemia	↑ free fraction of **phenytoin, valproate** (also carbamazepine, warfarin) → total level misleading → free level
Heart failure	↓ hepatic blood flow → ↑ **lidocaine, theophylline** • ↓ renal perfusion → ↑ digoxin, aminoglycosides
Pregnancy	↑ Vd, ↑ GFR, ↑ hepatic metabolism, ↓ albumin → ↓ levels of **lithium, lamotrigine, levetiracetam, phenytoin**, aminoglycosides • ↑ LMWH dose needs
Obesity	↑ Vd for lipophilic drugs; aminoglycosides dosed on **AdjBW** • vancomycin loading on actual weight
Critical illness	↑ Vd (fluids, capillary leak — aminoglycosides, vancomycin, β-lactams), **augmented renal clearance** (↓ levels), CRRT/ECMO changes
Age	Neonates: immature metabolism/renal function (↑ t½ of theophylline, gentamicin, phenobarbital) • elderly: ↓ renal/hepatic clearance, ↓ albumin
Smoking	**Induces CYP1A2 → ↓ theophylline, clozapine, olanzapine • stopping smoking → ↑ levels** (toxicity)
Genetics	**CYP2C9** (phenytoin, warfarin), **CYP2C19** (voriconazole), **CYP2D6** (TCAs, flecainide), **CYP3A5** (tacrolimus), **NAT2** acetylator (procainamide, isoniazid, hydralazine), **HLA-B*1502** (carbamazepine, phenytoin SJS)
Drug interactions	Enzyme **inducers** (rifampin, carbamazepine, phenytoin, phenobarbital, St John's wort) ↓ levels • **inhibitors** (azoles, macrolides, diltiazem, verapamil, amiodarone, ritonavir, cimetidine, fluvoxamine, ciprofloxacin, grapefruit) ↑ levels
Dialysis	Removes small, water-soluble, low-protein-bound drugs: **aminoglycosides, vancomycin (high-flux), lithium, valproate (partially), salicylates • not digoxin, amiodarone, phenytoin (largely)**

14. Worked Calculations (exam style)

Phenytoin albumin correction: measured phenytoin 8 mcg/mL, albumin 2.0 g/dL, normal renal function → C(corrected) = 8 / [(0.2 × 2.0) + 0.1] = 8 / 0.5 = **16 mcg/mL** → therapeutic (don't increase the dose!). If ESRD: 8 / [(0.1 × 2.0) + 0.1] = 8 / 0.3 = **26.7 mcg/mL** → toxic range.

Linear dose adjustment (e.g., lithium, digoxin, vancomycin trough approximation): lithium 0.5 mEq/L on 900 mg/day, target 0.8 → new dose = 900 × (0.8/0.5) = **1,440 mg/day** (≈ 1,500 mg/day) → recheck in 5 days. **Never use this proportional method for phenytoin** (nonlinear).

Aminoglycoside ke and half-life: gentamicin peak 8 mcg/mL (1 h after start), trough 1 mcg/mL 7 h later → ke = ln(8/1)/7 = 2.08/7 = **0.297 h^{-1}** → t½ = 0.693/0.297 = **2.3 h**. (Use to choose a new interval: time to fall from desired peak to desired trough = ln(Cpeak/Ctrough)/ke.)

Loading dose with salt factor: target theophylline 10 mcg/mL, Vd 0.5 L/kg, 70 kg, using aminophylline (S = 0.8), IV (F = 1) → LD = (0.5 × 70 × 10) / 0.8 = 350/0.8 = **437.5 mg aminophylline**. (If patient already has a level, LD = Vd × (Ctarget − Ccurrent) / S.)

Digoxin Fab dose (chronic toxicity): level 4 ng/mL, 80 kg → vials = (4 × 80)/100 = **3.2 → 4 vials**.

15. MASTER TABLE — All TDM Drugs

Drug	Therapeutic range	Sampling / timing	Toxicity & key factors
Gentamicin / tobramycin (conventional)	**Peak 5–10** (8–10 serious) • **trough < 1–2** mcg/mL	Peak 30 min after 30-min infusion • trough ≤ 30 min before dose • around 3rd dose	Nephro/ototoxicity (trough) • IBW/AdjBW • renal
Gentamicin / tobramycin (extended interval)	Trough undetectable (< 0.5–1)	Random level 6–14 h → **Hartford nomogram**	Avoid in pregnancy, burns, ascites, CrCl < 20, endocarditis
Gentamicin (synergy)	**Peak 3–4** • trough < 1	As above	Endocarditis
Amikacin	**Peak 20–35 • trough < 4–8** (extended: trough < 1–4)	As gentamicin	Same
Vancomycin	**AUC24 400–600** (MIC 1) • historical trough 10–15 / **15–20**	Bayesian (1–2 levels) or peak + trough • within 24–48 h • CI: Css 15–25	**AKI** (AUC > 600, + pip-tazo) • infusion reaction
Digoxin	**0.5–0.9 ng/mL (HF)** • ≤ 2 (AF)	**≥ 6–8 h post-dose** • SS 7–10 days (longer CKD)	GI, visual, arrhythmias • ↑ hypoK/hypoMg • amiodarone ↓ dose 50%
Phenytoin	**Total 10–20 • free 1–2** mcg/mL	Trough • 2 h after IV load • SS 5–7+ days	**Nonlinear** • albumin correction • nystagmus > 20, ataxia > 30, lethargy > 40
Fosphenytoin	As phenytoin	≥ 2 h after IV, ≥ 4 h after IM	Dosed in PE
Carbamazepine	**4–12** mcg/mL	Trough • **recheck at 3–4 weeks (autoinduction)**	Diplopia, ataxia, hyponatremia, SJS • 3A4
Valproic acid	**50–100** (mania 50–125) • free 5–15	Trough	Saturable binding → free level • hepatotoxicity, pancreatitis, thrombocytopenia, ↑ NH3 • carbapenems ↓
Phenobarbital	**15–40** mcg/mL	Trough • SS 2–3 weeks	Sedation, respiratory depression • inducer
Primidone	5–12 (+ PB 15–40)	Trough	Measure both
Ethosuximide	40–100 mcg/mL	Trough	GI
Lamotrigine / levetiracetam	~3–15 / ~12–46 (reference)	Trough	Pregnancy ↑ clearance • estrogen ↓ LTG • VPA ↑ LTG
Lithium	**Acute 0.8–1.2 • maintenance 0.6–1.0** • elderly 0.4–0.8 mEq/L	**12 h post-dose** • 5 days after change	> 1.5 tremor/GI, > 2 confusion/ataxia, > 2.5 seizures • ↑ thiazides, ACEi/ARB, NSAIDs, dehydration
Clozapine	**≥ 350 ng/mL**	12-h trough	Seizures • **smoking cessation ↑ levels** • fluvoxamine/cipro ↑ • ANC
Nortriptyline	**50–150 ng/mL** (window)	12-h trough at SS	QRS > 100 ms, anticholinergic • CYP2D6
Amitriptyline / imipramine	~80–200 / ~175–350 ng/mL (parent + metabolite)	Trough	Same
Theophylline	**5–15** (classic 10–20) mcg/mL	Trough / 4–6 h post SR	> 20 GI, tachycardia • > 30 seizures, arrhythmias • ↑ cipro, macrolides, smoking cessation; ↓ smoking, inducers

Drug	Therapeutic range	Sampling / timing	Toxicity & key factors
Tacrolimus	**~5–15 ng/mL** (whole blood trough)	Trough (12 h)	Nephro/neurotoxicity, hyperglycemia, ↑ K, ↓ Mg • 3A4/P-gp
Cyclosporine	**~100–400 ng/mL** (C0) or C2	Trough or 2 h	Nephrotoxicity, HTN, gingival hyperplasia, hirsutism
Sirolimus / everolimus	~5–15 / ~3–8 ng/mL	Trough	Hyperlipidemia, myelosuppression, wound healing
Lidocaine	**1.5–5** mcg/mL	During infusion	CNS toxicity > 5, seizures > 8 • ↑ HF/liver disease
Procainamide / NAPA	4–10 / 15–25 mcg/mL	Steady state	QT, torsades • NAPA renal • lupus
Quinidine / flecainide	2–5 / 0.2–1.0 mcg/mL	Trough	QT/proarrhythmia
Voriconazole	**Trough 1–5.5** mcg/mL	Day 5–7	Hallucinations, hepatotoxicity > 5.5 • CYP2C19
Posaconazole	Prophylaxis > 0.7 • treatment > 1–1.25	Trough day 5–7	Absorption issues
Itraconazole	Trough > 0.5–1	~2 weeks	Absorption
Flucytosine	**Peak 30–80** mcg/mL	2 h post-dose	> 100 marrow suppression • renal
Methotrexate (high-dose)	< 10 (24 h) • < 1 (48 h) • < 0.1 µmol/L (72 h)	24, 48, 72 h	Leucovorin, hydration, urine alkalinization • glucarpidase
Busulfan	Target AUC	Multiple samples after dose 1	VOD, seizures
Infliximab / adalimumab	Trough ≥ 5 / ≥ 7.5 mcg/mL	Trough + antibodies	IBD dose optimization
Warfarin	**INR 2–3** (2.5–3.5 mechanical mitral)	Regular INR	CYP2C9 DDIs, vitamin K
UFH	**aPTT 1.5–2.5×** / anti-Xa 0.3–0.7	6 h after change	HIT (platelets)
Enoxaparin	Anti-Xa 0.5–1.0 (q12h) / 1–2 (daily)	Peak 4 h after 3rd–4th dose	Renal, obesity, pregnancy
Salicylate	15–30 mg/dL (anti-inflammatory)	Repeat q2h in overdose	> 30–40 toxic • dialysis > 90–100 (acute)
Acetaminophen (overdose)	Nomogram line 150 mcg/mL at 4 h	4–24 h post-ingestion	NAC
Chloramphenicol	Peak 10–25 mcg/mL	Peak	Marrow suppression, gray baby

Exam Traps and Frequently Tested Points

1. **TDM criteria:** narrow therapeutic index, high PK variability, concentration–effect relationship, no easy clinical endpoint, assay available • treat the patient, not the level.
2. **Steady state = 4–5 half-lives** (loading dose gets you there faster) • **trough = just before next dose** • **$t_{½} = 0.693/ke$** • **$LD = Vd \times C / (S \times F)$**.
3. **Aminoglycosides:** concentration-dependent • peak efficacy, trough toxicity • gent/tobra **peak 5–10, trough < 1–2** • amikacin **peak 20–35, trough < 4–8** • synergy peak 3–4.
4. **Peak = 30 min after end of 30-min infusion** • extended-interval → **random level 6–14 h + Hartford nomogram** • dose on IBW (AdjBW if obese).
5. **Extended-interval aminoglycoside contraindicated:** pregnancy, burns, ascites, CrCl < 20, endocarditis synergy.
6. **Vancomycin: AUC 400–600** (Bayesian or 2 levels) • trough-only (15–20) no longer recommended • load 20–35 mg/kg • AKI ↑ with pip-tazo • oral vanco → no TDM.
7. **Digoxin 0.5–0.9 ng/mL in HF** • sample ≥ 6–8 h post-dose • hypoK/hypoMg/hyperCa ↑ toxicity • amiodarone → ↓ digoxin 50% • levels unreliable after DigiFab • DLIS in renal failure.

8. **Phenytoin 10–20 total / 1–2 free • nonlinear** (small dose ↑ → big level ↑) • **albumin correction C/(0.2 × alb + 0.1)** (0.1 × alb + 0.1 in ESRD).
9. **Phenytoin toxicity:** nystagmus > 20 → ataxia > 30 → lethargy > 40 • IV ≤ 50 mg/min (hypotension, arrhythmias) • fosphenytoin ≤ 150 mg PE/min • phenytoin sodium = 92% (S 0.92) • hold tube feeds.
10. **Carbamazepine 4–12 • autoinduction** → recheck at 3–4 weeks • hyponatremia, HLA-B*1502 • ↑ with macrolides, azoles, grapefruit.
11. **Valproate 50–100** (mania up to 125) • saturable protein binding → free level • thrombocytopenia, hyperammonemia • **carbapenems ↓ VPA**.
12. **Phenobarbital 15–40** (SS 2–3 weeks) • ethosuximide 40–100 • primidone 5–12 • lamotrigine/levetiracetam monitored in pregnancy.
13. **Lithium:** acute 0.8–1.2, maintenance 0.6–1.0, elderly 0.4–0.8 • **12 h post-dose** • 5 days after change • > 1.5 coarse tremor/GI, > 2 confusion/ataxia, > 2.5 seizures → dialysis.
14. **↑ Lithium:** thiazides, ACEi/ARB, NSAIDs, dehydration, low-Na diet, renal impairment • **↓ lithium:** caffeine, theophylline, pregnancy, high Na • monitor TSH, SCr, Ca.
15. **Clozapine ≥ 350 ng/mL • smoking cessation ↑ clozapine ~50%** (CYP1A2) • fluvoxamine/cipro ↑ • seizures dose-related • ANC monitoring.
16. **Nortriptyline 50–150 (therapeutic window)** • TCA toxicity → QRS > 100 ms → sodium bicarbonate • 2D6 inhibitors ↑ levels.
17. **Theophylline 5–15** (classic 10–20) • > 20 GI/tachycardia, > 30 seizures/arrhythmias • aminophylline = 80% theophylline • ↑ cipro, macrolides, cimetidine, fluvoxamine, HF, liver disease, quitting smoking • ↓ smoking, rifampin, phenytoin, carbamazepine.
18. **Tacrolimus 5–15, cyclosporine 100–400, sirolimus 5–15 ng/mL • whole blood trough** • CYP3A4/P-gp DDIs (azoles, macrolides, diltiazem/verapamil, PIs, grapefruit ↑; rifampin, carbamazepine, phenytoin, St John's wort ↓).
19. **Tacrolimus:** nephrotoxicity, tremor, hyperglycemia, hyperK, hypoMg • **cyclosporine:** + gingival hyperplasia, hirsutism, HTN, hyperlipidemia • **sirolimus:** hyperlipidemia, poor wound healing.
20. **Lidocaine 1.5–5** (CNS toxicity, seizures > 8; ↑ in HF/liver disease) • **procainamide 4–10 + NAPA 15–25** (NAPA renal; fast acetylators).
21. **Voriconazole trough 1–5.5** (CYP2C19; > 5.5 hallucinations/hepatotoxicity) • **posaconazole > 0.7 prophylaxis / > 1 treatment • flucytosine peak 30–80** (> 100 marrow).
22. **HDMTX:** levels 24/48/72 h (< 10 / < 1 / < 0.1 μM) → leucovorin until < 0.05–0.1 • hydration + urine pH ≥ 7 • avoid NSAIDs, PPIs, penicillins, TMP-SMX • glucarpidase.
23. **Anticoagulant monitoring:** warfarin INR 2–3 • UFH aPTT 1.5–2.5× or anti-Xa 0.3–0.7 • LMWH anti-Xa (peak 4 h, 0.5–1) only in renal/obesity/pregnancy • DOACs not routine.
24. **Acetaminophen nomogram:** level 4–24 h after acute ingestion, line 150 mcg/mL at 4 h → NAC • **salicylate** > 30 toxic; dialysis > 90–100.
25. **Free levels** for phenytoin and valproate in hypoalbuminemia, uremia, pregnancy, elderly, ICU.
26. **Smoking** induces CYP1A2 (theophylline, clozapine, olanzapine) — quitting ↑ levels • **never use proportional (linear) dose adjustment for phenytoin**.

> **Note:** therapeutic ranges vary slightly between references and laboratories. If your course slides or reference use different ranges, **use your course numbers for the exam**.